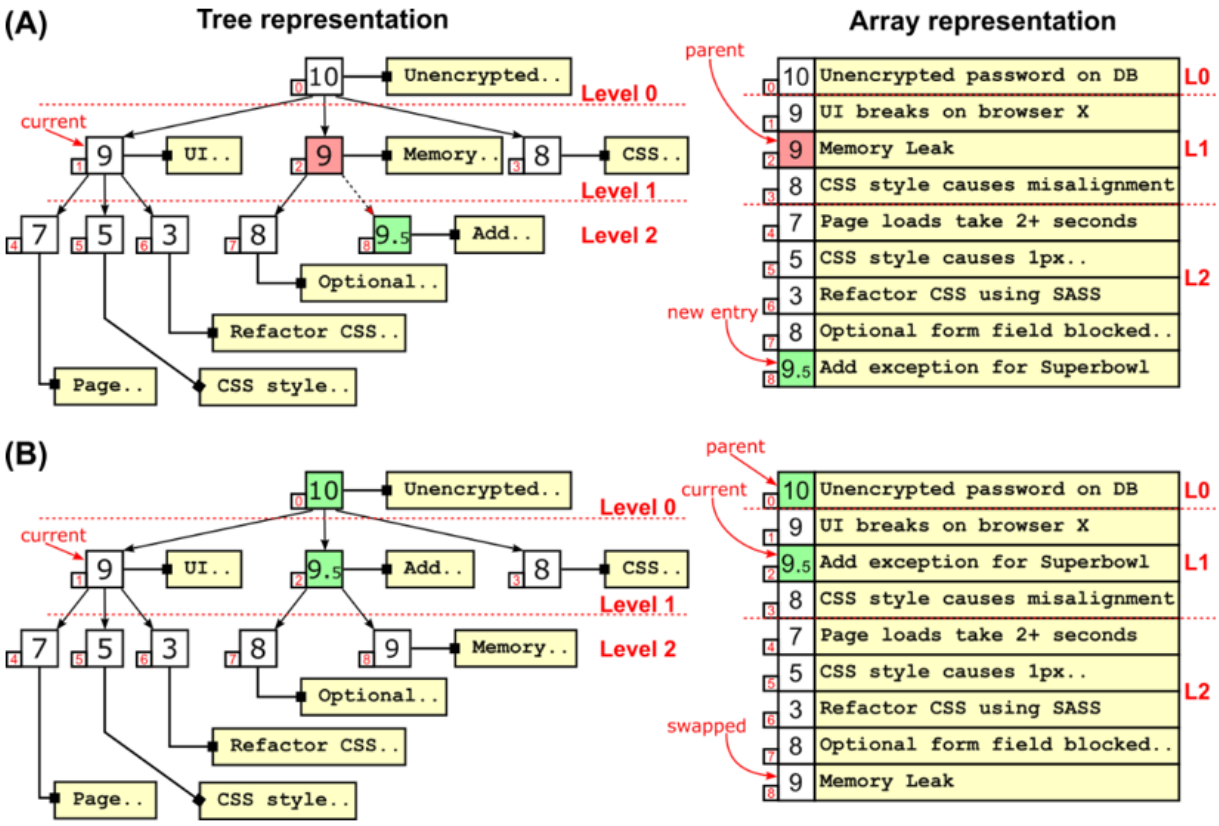



**Figure 2.9 Adding a new element to a ternary max-heap.**
**(A) The initial state of the heap; we first add the new element, with priority 9.5, at the end of the heap's array.**
**(B) Then we bubble it up till we find the right spot for it, i.e. the first spot where it does not violate the heap's properties.**

If we go back to our task management example, we can see how insert works on a ternary max-heap. Suppose we have the initial state shown in figure 2.9.A, describing the heap right after a call to `insert("Add exception for Superbowl", 9.5)` to add a particularly urgent task that marketing is pushing to have fixed by the need of the day